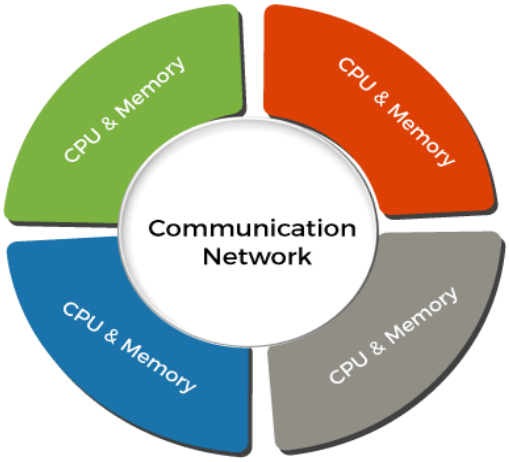
Distributed Operating System

A distributed operating system **(DOS)** is an essential type of operating system. Distributed systems use many central processors to serve multiple real-time applications and users. As a result, data processing jobs are distributed between the processors.

It connects multiple computers via a single communication channel. Furthermore, each of these systems has its own processor and memory. Additionally, these **CPUs** communicate via high-speed buses or telephone lines. Individual systems that communicate via a single channel are regarded as a single entity. They're also known as **loosely coupled systems**.



This operating system

consists of numerous computers, nodes, and sites joined together via **LAN/WAN** lines. It enables the distribution of full systems on a couple of center processors, and it supports many real-time products and different users. Distributed operating systems can share their computing resources and I/O files while providing users with virtual machine abstraction.

Types of Distributed Operating System

There are various types of Distributed Operating systems. Some of them are as follows:

1. **Client-Server Systems**
2. **Peer-to-Peer Systems**
3. **Middleware**
4. **Three-tier**
5. **N-tier**

Client-Server System

This type of system requires the client to request a resource, after which the server gives the requested resource. When a client connects to a server, the server may serve multiple clients at the same time.

Client-Server Systems are also referred to as "Tightly Coupled Operating Systems". This system is primarily intended for multiprocessors and homogenous multicomputer. Client-Server Systems function as a centralized server since they approve all requests issued by client systems.

Server systems can be divided into two parts:

1. Computer Server System

This system allows the interface, and the client then sends its own requests to be executed as an action. After completing the activity, it sends a back response and transfers the result to the client.

2. File Server System

It provides a file system interface for clients, allowing them to execute actions like file creation, updating, deletion, and more.

Peer-to-Peer System

The nodes play an important role in this system. The task is evenly distributed among the nodes. Additionally, these nodes can share data and resources as needed. Once again, they require a network to connect.

The Peer-to-Peer System is known as a "Loosely Couple System". This concept is used in computer network applications since they contain a large number of processors that do not share memory or clocks. Each processor has its own local memory, and they interact with one another via a variety of communication methods like telephone lines or high-speed buses.

Middleware

Middleware enables the interoperability of all applications running on different operating systems. Those programs are capable of transferring all data to one other by using these services.

Three-tier

The information about the client is saved in the intermediate tier rather than in the client, which simplifies development. This type of architecture is most commonly used in online applications.

N-tier

When a server or application has to transmit requests to other enterprise services on the network, n-tier systems are used.

Features of Distributed Operating System

There are various features of the distributed operating system. Some of them are as follows:

Openness

It means that the system's services are freely displayed through interfaces. Furthermore, these interfaces only give the service syntax. For example, the type of function, its return type, parameters, and so on. Interface Definition Languages are used to create these interfaces (IDL).

Scalability

It refers to the fact that the system's efficiency should not vary as new nodes are added to the system. Furthermore, the performance of a system with 100 nodes should be the same as that of a system with 1000 nodes.

Resource Sharing

Its most essential feature is that it allows users to share resources. They can also share resources in a secure and controlled manner. Printers, files, data, storage, web pages, etc., are examples of shared resources.

Flexibility

A DOS's flexibility is enhanced by modular qualities and delivers a more advanced range of high-level services. The kernel/ microkernel's quality and completeness simplify the implementation of such services.

Transparency

It is the most important feature of the distributed operating system. The primary purpose of a distributed operating system is to hide the fact that resources are shared. Transparency also implies that the user should be unaware that the resources he is accessing are shared. Furthermore, the system should be a separate independent unit for the user.

Heterogeneity

The components of distributed systems may differ and vary in operating systems, networks, programming languages, computer hardware, and implementations by different developers.

Fault Tolerance

Fault tolerance is that process in which user may continue their work if the software or hardware fails.

Examples of Distributed Operating System

There are various examples of the distributed operating system. Some of them are as follows:

Solaris

It is designed for the SUN multiprocessor workstations

Locus

It may be accessed local and remote files at the same time without any location hindrance.

Mach

It allows the multithreading and multitasking features.

Applications of Distributed Operating System

There are various applications of the distributed operating system. Some of them are as follows:

Network Applications

DOS is used by many network applications, including the Web, peer-to-peer networks, multiplayer web-based games, and virtual communities.

Telecommunication Networks

DOS is useful in phones and cellular networks. A DOS can be found in networks like the Internet, wireless sensor networks, and routing algorithms.

Parallel Computation

DOS is the basis of systematic computing, which includes cluster computing and grid computing, and a variety of volunteer computing projects.

Real-Time Process Control

The real-time process control system operates with a deadline, and such examples include aircraft control systems.

Advantages and Disadvantages of Distributed Operating System

There are various advantages and disadvantages of the distributed operating system. Some of them are as follows:

Advantages

There are various advantages of the distributed operating system. Some of them are as follow:

1. It may share all resources (CPU, disk, network interface, nodes, computers, and so on) from one site to another, increasing data availability across the entire system.
2. It reduces the probability of data corruption because all data is replicated across all sites; if one site fails, the user can access data from another operational site.

3. The entire system operates independently of one another, and as a result, if one site crashes, the entire system does not halt.
4. It increases the speed of data exchange from one site to another site.
5. It is an open system since it may be accessed from both local and remote locations.
6. It helps in the reduction of data processing time.
7. Most distributed systems are made up of several nodes that interact to make them fault-tolerant. If a single machine fails, the system remains operational.

Disadvantages

There are various disadvantages of the distributed operating system. Some of them are as follows:

1. The system must decide which jobs must be executed when they must be executed, and where they must be executed. A scheduler has limitations, which can lead to underutilized hardware and unpredictable runtimes.
2. It is hard to implement adequate security in DOS since the nodes and connections must be secured.
3. The database connected to a DOS is relatively complicated and hard to manage in contrast to a single-user system.
4. The underlying software is extremely complex and is not understood very well compared to other systems.
5. The more widely distributed a system is, the more communication latency can be expected. As a result, teams and developers must choose between availability, consistency, and latency.
6. These systems aren't widely available because they're thought to be too expensive.
7. Gathering, processing, presenting, and monitoring hardware use metrics for big clusters can be a real issue.

Linux os

Linux is an open-source operating system like other operating systems such as Microsoft Windows

, Apple Mac OS, iOS, Google android, etc. An operating system is a software that enables the communication between computer hardware and software. It conveys input to get processed by the processor and brings output to the hardware to display it. This is the basic function of an operating system. Although it performs many other important tasks, let's not talk about that.

Linux is around us since the mid-90s. It can be used from wristwatches to supercomputers. It is everywhere in our phones, laptops, PCs, cars and even in refrigerators. It is very much famous among developers and normal computer users.

Evolution of Linux OS

The Linux OS

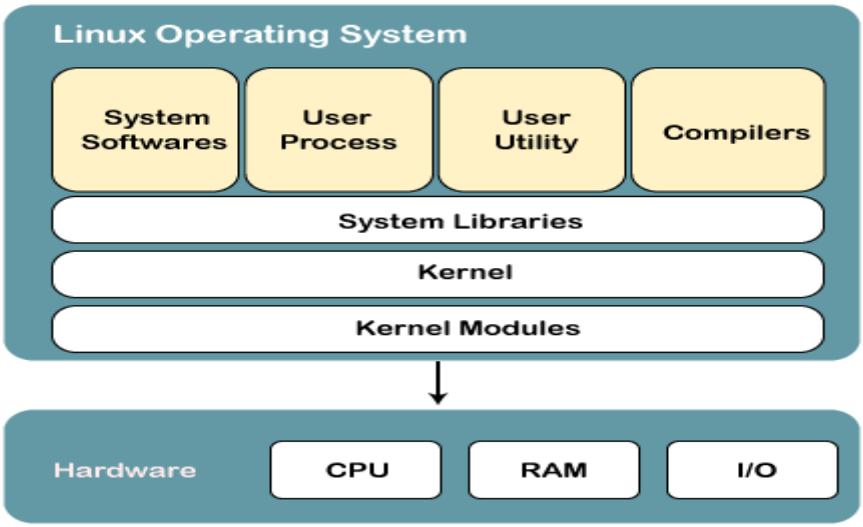
was developed by **Linus Torvalds** in **1991**, which sprouted as an idea to improve the UNIX OS. He suggested improvements but was rejected by UNIX designers. Therefore, he thought of launching an OS, designed in a way that could be modified by its users.

Nowadays, Linux is the fastest-growing OS. It is used from phones to supercomputers by almost all major hardware devices.

Structure Of Linux Operating System

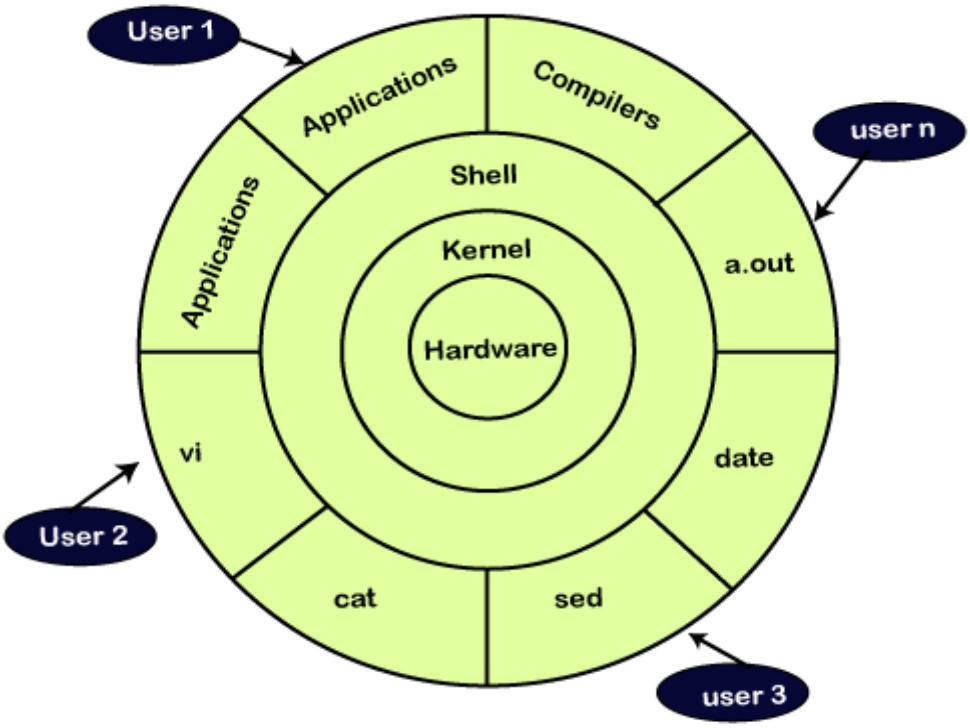
An operating system is a collection of software, each designed for a specific function.

Linux OS has following components:



1) Kernel

Linux kernel is the core part of the operating system. It establishes communication between devices and software. Moreover, it manages system resources. It has four responsibilities:



- **device management:** A system has many devices connected to it like CPU, a memory device, sound cards, graphic cards, etc. A kernel stores all the data related to all the devices in the device driver (without this kernel won't be able to control the devices). Thus kernel knows what a device can do and how to manipulate it to bring out the best performance. It also manages communication between all the devices. The kernel has certain rules that have to be followed by all the devices.
- **Memory management:** Another function that kernel has to manage is the memory management. The kernel keeps track of used and unused memory and makes sure that processes shouldn't manipulate data of each other using virtual memory addresses.
- **Process management:** In the process, management kernel assigns enough time and gives priorities to processes before handling CPU to other processes. It also deals with security and ownership information.
- **Handling system calls:** Handling system calls means a programmer can write a query or ask the kernel to perform a task.

2) System Libraries

System libraries are special programs that help in accessing the kernel's features. A kernel has to be triggered to perform a task, and this triggering is done by the applications. But applications must know how to place a system call because each kernel has a different set of system calls. Programmers have developed a standard library of procedures to communicate with the kernel. Each operating system supports these standards, and then these are transferred to system calls for that operating system.

The most well-known system library for Linux is Glibc (GNU C library).

3) System Tools

Linux OS has a set of utility tools, which are usually simple commands. It is a software which GNU project has written and publish under their open source license so that software is freely available to everyone.

With the help of commands, you can access your files, edit and manipulate data in your directories or files, change the location of files, or anything.

4) Development Tools

With the above three components, your OS is running and working. But to update your system, you have additional tools and libraries. These additional tools and libraries are written by the programmers and are called toolchain. A toolchain is a vital development tool used by the developers to produce a working application.

5) End User Tools

These end tools make a system unique for a user. End tools are not required for the operating system but are necessary for a user.

Some examples of end tools are graphic design tools, office suites, browsers, multimedia players, etc.

Why use Linux?

This is one of the most asked questions about Linux systems. Why do we use a different and bit complex operating system, if we have a simple operating system like Windows? So there are various features of Linux

systems that make it completely different and one of the most used operating systems. Linux may be a perfect operating system *if you want to get rid of viruses, malware, slowdowns, crashes, costly repairs,* and many more. Further, it provides various advantages over other operating systems

, and we don't have to pay for it. Let's have a look at some of its special features that will attract you to switch your operating system.

Free & Open Source Operating System

Most OS come in a compiled format means the main source code has run through a program called a compiler that translates the source code into a language that is known to the computer.

Modifying this compiled code is a tough job.

On the other hand, open-source is completely different. The source code is included with the compiled version and allows modification by anyone having some knowledge. It gives us the freedom to run the program, freedom to change the code according to our use, freedom to redistribute its copies, and freedom to distribute copies, which are modified by us.

*In short, Linux is an operating system that is "**for the people, by the people**."*

And we can dive in Linux without paying any cost. We can install it on Multiple machines without paying any cost.

It is secure

Linux supports various security options that will save you from viruses, malware, slowdowns, crashes. Further, it will keep your data protected. Its security feature is the main reason that it is the most favorable option for developers. It is not completely safe, but it is less vulnerable than others. Each application needs to authorize by the admin user. The virus cannot be executed until the administrator provides the access password. Linux systems do not require any antivirus program.

Favorable choice of Developers

Linux is suitable for the developers, as it supports almost all of the most used programming languages such as C ,/C++, , Java, , Python ,, Ruby

, and more. Further, it facilitates with a vast range of useful applications for development.

Developers find that the Linux terminal is much better than the Windows command line, So, they prefer terminal over the Windows command line. The package manager on Linux system helps programmers to understand how things are done. Bash scripting

is also a functional feature for the programmers. Also, the SSH support helps to manage the servers quickly.

A flexible operating system

Linux is a flexible OS, as, it can be used for desktop applications, embedded systems, and server applications. It can be used from wristwatches to supercomputers. It is everywhere in our phones, laptops, PCs, cars and even in refrigerators. Further, it supports various customization options.

Linux Distributions

Many agencies modified the Linux operating system and makes their Linux distributions. There are many Linux distributions available in the market. It provides a different flavor of the Linux operating system to the users. We can choose any distribution according to our needs. Some popular distros are Ubuntu, Fedora, Debian, Linux Mint, Arch Linux, and many more.

For the beginners, Ubuntu and Linux Mint are considered useful and, for the proficient developer, Debian and Fedora would be a good choice. To Get a list of distributions, visit Linux Distributions

.

How does Linux work?

Linux is a UNIX-like

operating system, but it supports a range of hardware devices from phones to supercomputers. Every Linux-based operating system has the Linux kernel and set of software packages to manage hardware resources.

Also, Linux OS includes some core GNU tools to provide a way to manage the kernel resources, install software, configure the security setting and performance, and many more. All these tools are packaged together to make a functional operating system.

Advantages of Linux

Linux is an open-source operating system like Windows and MacOS. It is not just limited to the operating system, but nowadays, it is also used as a platform to run **desktops, servers, and embedded systems**. It provides various distributions and variations as it is open source and has a modular design. The **kernel** *is a core part of* the Linux system.

Linux system is used to manage various services such as **process scheduling, application scheduling, basic peripheral devices, file system, and more**. Linux provides various advantages over other operating systems such as Windows and macOS. So, it is used in almost every field, from cars to home appliances and smartphones to servers (supercomputers).

In this section, we will see some major advantages of the Linux system. Further, we will see the advantages of Linux over other operating systems and will determine why it is better than other operating systems.

Why is Linux better than other operating systems?

There are many features of the Linux operating system that demonstrate that it is better than other operating systems. However, in some prospective other operating systems can be more useful than Linux. Let's see the top 20 advantages of Linux OS.

Advantages of Linux

1. pen Source

As it is open-source, its source code is easily available. Anyone having programming knowledge can customize the operating system. One can contribute, modify, distribute, and enhance the code for any purpose.

2. Security

The Linux security feature is the main reason that it is the most favorable option for developers. It is not completely safe, but it is less vulnerable than others. Each application needs to authorize by the admin user. The virus is not executed until the administrator provides the access password. Linux systems do not require any antivirus program.

3. Free

Certainly, the biggest advantage of the Linux system is that it is free to use. We can easily download it, and there is no need to buy the license for it. It is distributed under GNU GPL (General Public License). Comparatively, we have to pay a huge amount for the license of the other operating systems.

4. Lightweight

Linux is lightweight. The requirements for running Linux are much less than other operating systems. In Linux, the memory footprint and disk space are also lower. Generally, most of the Linux distributions required as little as 128MB of RAM around the same amount for disk space.

5. Stability

Linux is more stable than other operating systems. Linux does not require to reboot the system to maintain performance levels. It rarely hangs up or slow down. It has big up-times.

6. Performance

Linux system provides high performance over different networks. It is capable of handling a large number of users simultaneously.

7. Flexibility

Linux operating system is very flexible. It can be used for desktop applications, embedded systems, and server applications too. It also provides various restriction options for specific computers. We can install only necessary components for a system

8. Software Updates

In Linux, the software updates are in user control. We can select the required updates. There a large number of system updates are available. These updates are much faster than other operating systems. So, the system updates can be installed easily without facing any issue.

9. Distributions/ Distros

There are many Linux distributions available in the market. It provides various options and flavors of Linux to the users. We can choose any distros according to our needs. Some popular distros are **Ubuntu, Fedora, Debian, Linux Mint, Arch Linux,** and many more.

For the beginners, Ubuntu and Linux Mint would be useful and, Debian and Fedora would be good choices for proficient programmers.

10. Live CD/USB

Almost all Linux distributions have **a Live CD/USB** option. It allows us to try or run the Linux operating system without installing it.

11. Graphical User Interface

Linux is a command-line based OS but, it provides an interactive user interface like Windows.

12. Suitable for programmers

It supports almost all of the most used programming languages such as C/C++, Java, Python, Ruby, and more. Further, it offers a vast range of useful applications for development.

The programmers prefer the Linux terminal over the Windows command line. The package manager on Linux system helps programmers to understand how things are done. Bash scripting is also a functional feature for the programmers. It also provides support for SSH, which helps in managing the servers quickly.

13. Community Support

Linux provides large community support. We can find support from various sources. There are many forums available on the web to assist users. Further, developers from the various opensource communities are ready to help us.

14. Privacy

Linux always takes care of user privacy as it never takes much private data from the user. Comparatively, other operating systems ask for the user's private data.

15. Networking

Linux facilitates with powerful support for networking. The client-server systems can be easily set to a Linux system. It provides various command-line tools such as ssh, ip, mail, telnet, and more for connectivity with the other systems and servers. Tasks such as network backup are much faster than others.

16. Compatibility

Linux is compatible with a large number of file formats as it supports almost all file formats.

17. Installation

Linux installation process takes less time than other operating systems such as Windows. Further, its installation process is much easy as it requires less user input. It does not require much more system configuration even it can be easily installed on old machines having less configuration.

18. Multiple Desktop Support

Linux system provides multiple desktop environment support for its enhanced use. The desktop environment option can be selected during installation. We can select any desktop environment such as **GNOME (GNU Network Object Model Environment)** or **KDE (K Desktop Environment)** as both have their specific environment.

19. Multitasking

It is a multitasking operating system as it can run multiple tasks simultaneously without affecting the system speed.

20. Heavily Documented for beginners

There are many command-line options that provide documentation on commands, libraries, standards such as manual pages and info pages. Also, there are plenty of documents available on the internet in different formats, such as Linux tutorials, Linux documentation project, Serverfault, and more. To help the beginners, several communities are available such as **Ask Ubuntu**, Reddit, and **StackOverflow.**

UNIX operating system

UNIX is a powerful Operating System initially developed by Ken Thompson, Dennis Ritchie at AT&T Bell laboratories in 1970. It is prevalent among scientific, engineering, and academic institutions due to its most appreciative features like multitasking, flexibility, and many more. In UNIX, the file system is a hierarchical structure of files and directories where users can store and retrieve information using the files.

Features of UNIX Operating System:

Multitasking: A UNIX operating system is a multitasking operating system that allows you to initiate more than one task from the same terminal so that one task is performed as a foreground and the other task as a background process.

Multi-user: UNIX operating system supports more than one user to access computer resources like main memory, hard disk, tape drives, etc. Multiple users can log on to the system from different terminals and run different jobs that share the resources of a command terminal. It deals with the principle of time-sharing. Time-sharing is done by a scheduler that divides the CPU time into several segments also called a time slice, and each segment is assigned to each user on a scheduled basis. This time slice is tiny. When this time is expired, it passes control to the following user on the system. Each user executes their set of instructions within their time slice.

Portability: This feature makes the UNIX work on different machines and platforms with the easy transfer of code to any computer system. Since a significant portion of UNIX is written in C language, and only a tiny portion is coded in assembly language for specific hardware.

File Security and Protection: Being a multi-user system, UNIX makes special consideration for file and system security. UNIX has different levels of security using assigning username and password to individual users ensuring the authentication, at the level providing file access permission viz. read, write and execute and lastly file encryption to change the file into an unreadable format.

Command Structure: UNIX commands are easy to understand and simple to use. Example: "cp", mv etc. While working in the UNIX environment, the UNIX commands are case-sensitive and are entered in lower case.

Communication: In UNIX, communication is an excellent feature that enables the user to communicate worldwide. It supports various communication facilities provided using the write command, mail command, talk command, etc.

Open Source: UNIX operating system is open source it means it is freely available to all and is a community-based development project.

Accounting: UNIX keeps an account of jobs created by the user. This feature enhances the system performance in terms of CPU monitoring and disk space checking. It allows you to keep an account of disk space used by each user, and the disk space can be limited by each other. You can assign every user a different disk quota. The root user can perform these accounting tasks using various commands such as quota, df, du, etc.

UNIX Tools and Utilities: UNIX system provides various types of tools and utilities facilities such as UNIX grep, sed and awk, etc. Some of the general-purpose tools are compilers, interpreters, network applications, etc. It also includes various server programs which provide remote and administration services.

The structure of Unix OS Layers are as follows:

While working with UNIX OS, several layers of this system provide interaction between the pc hardware and the user. Following is the description of each and every layer structure in UNIX system:

Layer-1: Hardware -

This layer of UNIX consists of all hardware-related information in the UNIX environment.

Layer-2: Kernel -

The core of the operating system that's liable for maintaining the full functionality is named the kernel. The kernel of UNIX runs on the particular machine hardware and interacts with the hardware effectively.

It also works as a device manager and performs valuable functions for the processes which require access to the peripheral devices connected to the computer. The kernel controls these devices through device drivers.

The kernel also manages the memory. Processes are executed programs that have owner's humans or systems who initiate their execution.

The system must provide all processes with access to an adequate amount of memory, and a few processes require a lot of it. To make effective use of main memory and to allocate a sufficient amount of memory to every process. It uses essential techniques like paging, swapping, and virtual storage.

Layer-3: The Shell -

The Shell is an interpreter that interprets the command submitted by the user at the terminal, and calls the program you simply want.

It also keeps a history of the list of the commands you have typed in. If you need to repeat a command you typed it, use the cursor keys to scroll up and down the list or type history for a list of previous commands. There are various commands like cat, mv, cat, grep, id, wc, and many more.

Types of Shell in UNIX System:

- **Bourne Shell:** This Shell is simply called the Shell. It was the first Shell for UNIX OS. It is still the most widely available Shell on a UNIX system.
- **C Shell:** The C shell is another popular shell commonly available on a UNIX system. The C shell was developed by the University of California at Berkeley and removed some of the shortcomings of the Bourne shell.
- **Korn Shell:** This Shell was created by David Korn to address the Bourne Shell's user-interaction issues and to deal with the shortcomings of the C shell's scripting quirks.

Layer-4: Application Programs Layer -

It is the outermost layer that executes the given external applications. UNIX distributions typically come with several useful applications programs as standard. **For Example:** emacs editor, StarOffice, xv image viewer, g++ compiler etc.